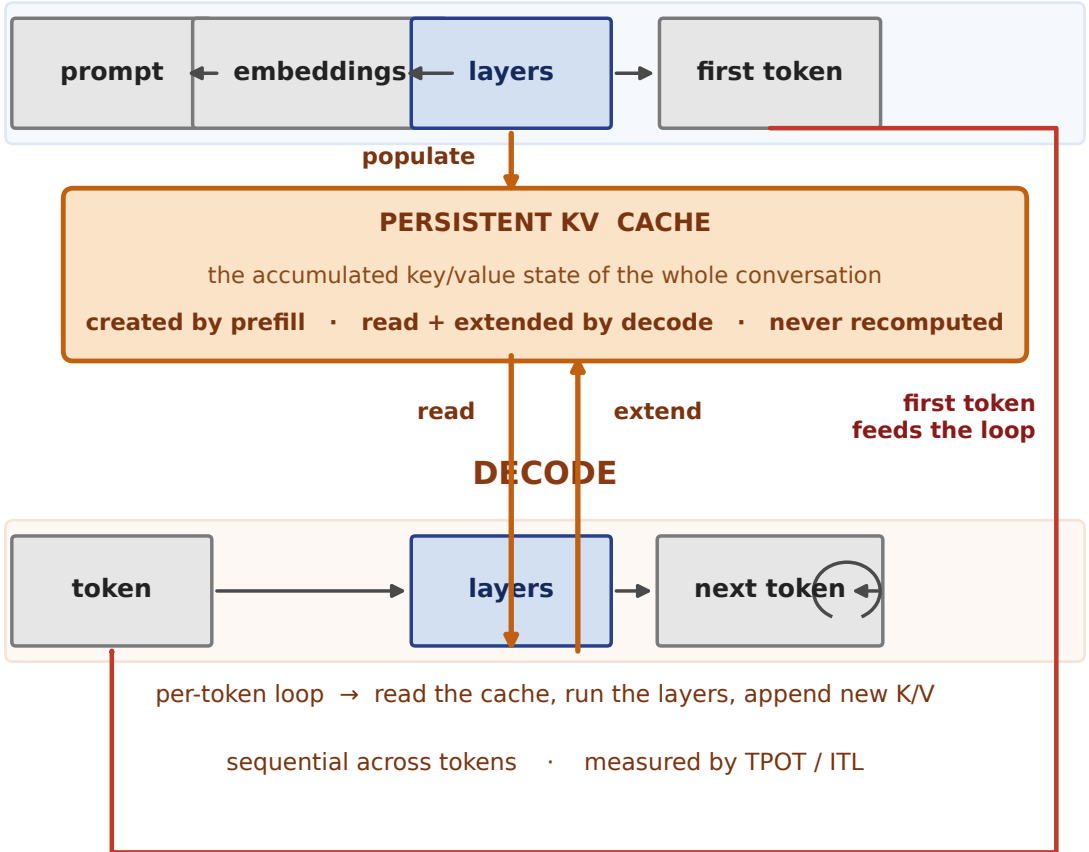


PREFILL

one-shot pass over the whole prompt → populate the cache → emit the first token

parallel across prompt positions · measured by TTFT



compute- vs bandwidth-bound is an operating-point property

a property of the model × workload × kernel × hardware — not a fixed rule

Prefill: typically higher arithmetic intensity

Decode: often bandwidth-sensitive at low batch